

1. Authentication Service

Database Philosophy: This database is highly secure and specialized. It only stores information required to authenticate a user and manage security credentials. It does not store personal info like names or preferences.

Table: Users (Manages core identity)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
Email	string	User's unique login email (Indexed, Unique, VarChar(255))
PasswordHash	string	Cryptographically hashed and salted password string (VarChar(512))
IsLockedOut	bool	Safety flag tracking brute-force states
LockedUntil	DateTime?	Expiration timestamp of the lockout penalty window (Nullable)
CreatedAt	DateTime	Record initialization timestamp

Table: RefreshTokens (Stores active tokens for sliding rotation sessions)

Column Name	Type	Description / Example
Id	int	Primary Key
UserId	int	Foreign Key linking directly to Users.Id
Token	string	Unique high-entropy token string (Unique Index, VarChar(512))
ExpiresAt	DateTime	When the token is no longer valid
CreatedAt	DateTime	Creation log timestamp
RevokedAt	DateTime?	Cancellation timestamp if explicitly invalidated (Nullable)

Table: OtpCodes (Stores temporary verification factors)

Column Name	Type	Description / Example
Id	int	Primary Key
Email	string	The email destination the code challenge was dispatched to (Indexed)
Code	string	The 6-digit hashed verification factor
ExpiresAt	DateTime	Strict expiration boundary (Exactly 10 minutes from creation)
IsUsed	bool	Single-use flag preventing replay vectors

Table: LoginAttempts (Audits consecutive authorization failures)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
Email	string	Inbound email address submitted during authorization try (VarChar(255))
AttemptedAt	DateTime	High-precision event tracking timestamp
IsSuccess	bool	Boolean verification pass indicator
IpAddress	string	Client IP tracker variable (VarChar(45))

2. User Profile Service

Database Philosophy: This database stores all personal metadata and user-facing layout configurations. It links to the Authentication layer via a shared identifier (`UserId` used as primary identity).

Table: UserProfiles (Manages personal metadata)

Column Name	Type	Description / Example
Id	int	Primary Key (Must be the same as <code>Users.Id</code> from Auth database)
FirstName	string	User's first name, e.g., "Ahmed"
LastName	string	User's last name, e.g., "Mohamed"
Email	string	Denormalized replication for user display, e.g., "ahmed@example.com"
PhoneNumber	string	User contact phone number, e.g., "+201234567890"

Column Name	Type	Description / Example
ProfilePictureUrl	string	Location path string of image binary layout, e.g., "/images/profiles/1.jpg"
IsPremiumCached	bool	Local replica cache flag synced asynchronously from Subscription Microservice events
MemberSince	DateTime	Application profile creation milestone date

Table: UserSettings (Manages user localization preferences)

Column Name	Type	Description / Example
Id	int	Primary Key (Must match UserProfiles.Id cleanly)
Language	string	System localization code, e.g., "en"
Theme	string	Visual display configuration variable, e.g., "dark"
NotificationSettings	JSON/string	Toggles mapping tracking targets, e.g., { "workoutReminders": true, "mealReminders": true }
PrivacySettings	JSON/string	Visibility access map modifiers, e.g., { "profileVisibility": "private" }
UnitSettings	JSON/string	Metric configuration metrics constraints, e.g., { "weight": "kg", "height": "cm" }

3. Fitness Calculation Engine (FCE)

Database Philosophy: This represents the algorithmic "brains" of the system. It stores physical biometrics logs, holds target computed scientific variables, and runs matching configuration tables to output structural tracks.

Table: UserFitnessStats (Stores raw physical input variables)

Column Name	Type	Description / Example
Id	int	Primary Key (Must match Users.Id from Auth database)
Weight	double	Current recorded mass scale in kilograms, e.g., 75.5
Height	double	Current recorded vertical scaling configuration in centimeters, e.g., 175
Age	int	Current age parameter constraint, e.g., 25

Column Name	Type	Description / Example
Gender	string	Biological indicator checked during formulas execution, e.g., "Male"
Goal	string	Focus objective vector selected by the user, e.g., "Lose Weight"
ActivityLevel	string	Multiplier scale mapping, e.g., "Intermediate"
RecordedAt	DateTime	Explicit entry initialization timestamp

Table: CalculatedMetrics (Stores derived metabolic outputs)

Column Name	Type	Description / Example
Id	int	Primary Key (Must match Users.Id from Auth database)
BMR	double	Derived Basal Metabolic Rate score calculation, e.g., 1728.75
TDEE	double	Derived Total Daily Energy Expenditure value, e.g., 2679.56
CalorieTarget	double	Final system calculated target caloric allocation balance, e.g., 2179.56
Status	string	Evaluated classification tier baseline category, e.g., "Normal" (Weak / Normal / Hard)
CalculatedAt	DateTime	Explicit execution calculation completion timestamp

Table: FitnessPlanConfigs (Rulebook blueprint configurations)

Column Name	Type	Description / Example
Id	int	Primary Key identifier
Goal	string	Associated objective classifier pattern target, e.g., "Lose Weight"
Status	string	Associated calorie tier status constraint variable, e.g., "Normal"
CalorieMin	double	Minimum allocation threshold evaluation, e.g., 1801
CalorieMax	double	Maximum boundary constraints alignment, e.g., 2500
ExternalPlanId	string	Direct structural key pairing across external content libraries, e.g., "plan_lw_normal"
PlanName	string	Human-readable descriptor string, e.g., "Weight Loss - Normal Intensity"

Column Name	Type	Description / Example
WorkoutsPerWeek	int	Training day frequency per schedule, e.g., 4

Table: UserAssignedPlans (Tracks active strategy assignments)

Column Name	Type	Description / Example
Id	int	Primary Key (Must match <code>Users.Id</code> from Auth database)
ExternalPlanId	string	Connected strategy target config key identifier, e.g., "plan_lw_normal"
AssignedAt	DateTime	System strategy allocation timestamp
IsActive	bool	Execution tracker flag denoting current status

Table: UserPlanHistory (Immutable tracking reassignment audit logs)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
UserId	int	Target user index identifier link
ExternalPlanId	string	Past active plan identifier record
AssignedAt	DateTime	Initial implementation timeline milestone
EndedAt	DateTime?	Termination tracking timestamp if recalculated (Nullable)
ReasonForChange	string	Action description context tracker, e.g., "weight_update"

4. Workout Service 💪

Database Philosophy: This represents the structural product content library. It houses blueprint templates for routine structures, exercise specifications arrays, and logs execution session initializations.

Table: WorkoutPlans (Top-level plan definitions)

Column Name	Type	Description / Example
Id	int	Primary Key

Column Name	Type	Description / Example
ExternalPlanId	string	Indexed, Unique mapping code paired directly across FCE logic, e.g., "plan_lw_normal"
Name	string	Title name string, e.g., "Weight Loss – Normal Intensity"
Description	string	Structural overview description block, e.g., "Balanced workout plan for weight loss"
Goal	string	Focus requirement string, e.g., "Lose Weight"
Status	string	Caloric tier mapping descriptor, e.g., "Normal"
Difficulty	string	Enforced scale classifier target, e.g., "Intermediate"

Table: Workouts (Specific training routine modules)

Column Name	Type	Description / Example
Id	int	Primary Key
WorkoutPlanId	int	Foreign Key mapping directly back to WorkoutPlans.Id
Name	string	Title layout label descriptor, e.g., "Full Body Strength"
Category	string	Index target tag classification grouping, e.g., "full-body"
Difficulty	string	Routine capability ranking target, e.g., "Intermediate"
DurationInMinutes	int	Target expected window scale allocation, e.g., 45
CaloriesBurn	int	Mathematically computed estimated output balance, e.g., 350
ImageUrl	string	Image location file path destination string, e.g., "/images/workouts/fb.jpg"
IsPremium	bool	Subscription gate flag protecting access paths

Table: Exercises (Master movement template definitions)

Column Name	Type	Description / Example
Id	int	Primary Key
Name	string	Clean movement name key, e.g., "Push-ups"
Description	string	Technical execution text instruction guidelines, e.g., "A calisthenic exercise..."
VideoUrl	string	Media storage loop file location string, e.g., "/videos/exercises/pushup.mp4"
TargetMuscles	JSON/string	String collection list arrays denoting targets, e.g., ["Chest", "Triceps"]

Column Name	Type	Description / Example
EquipmentNeeded	JSON/string	Structural gear requirement identifiers list, e.g., ["Bodyweight"]
Difficulty	string	Classification scale, e.g., "Beginner"

Table: WorkoutExercises (Template mapping sequence join table)

Column Name	Type	Description / Example
Id	int	Primary Key
WorkoutId	int	Foreign Key referencing parent routine identifier <code>Workouts.Id</code>
ExerciseId	int	Foreign Key referencing specific move index <code>Exercises.Id</code>
OrderIndex	int	Sequence positioning sorting metric configuration, e.g., 1
SetsDefault	int	Baseline training structure volume tracking variable, e.g., 3
RepsDefault	string	Default target iteration count layout format, e.g., "12-15"
RestTimeInSeconds	int	Target standard resting space constraint interval, e.g., 60

Table: WorkoutSessions (Isolated runtime track session instances)

Column Name	Type	Description / Example
SessionId	string	Globally unique random token tracking identifier (Primary Key)
UserId	int	Target user index identifier link (Indexed)
WorkoutId	int	Foreign Key matching target training reference row <code>Workouts.Id</code>
StartedAt	DateTime	Operational state runtime initiation timestamp
Status	string	Session management control status flag (Active , Completed , Abandoned)

5. Nutrition Service 🥗

Database Philosophy: This microservice acts as the programmatic repository for all recipe matrices, dietary structures, and caloric grouping frameworks.

Table: MealPlans (Target grouping frameworks configurations)

Column Name	Type	Description / Example
Id	int	Primary Key
Name	string	Menu categorization theme string, e.g., "High Protein 2200 kcal"
Description	string	Objective summary statement text block, e.g., "A plan for muscle gain."
CalorieTarget	int	Central energy property baseline target calculation index, e.g., 2200

Table: Meals (Specific culinary item profiles)

Column Name	Type	Description / Example
Id	int	Primary Key identifier
MealPlanId	int	Foreign Key matching back to MealPlans.Id
Name	string	Culinary label description name key, e.g., "Grilled Chicken Lunch"
MealType	string	Specific distribution slot allocation tier, e.g., "Lunch"
PrepTimeInMinutes	int	Operational preparation runtime limit target, e.g., 30
Difficulty	string	Execution capability rating marker, e.g., "Medium"
ImageUrl	string	Photo vector path asset directory pointer string, e.g., "/images/meals/1.jpg"
IngredientsJson	string	Raw text string details identifying absolute asset ratios
InstructionsJson	string	Text arrays detailing step-by-step preparation guidelines
VariationsJson	string?	Custom alternatives for adaptation paths (Nullable)
AllergensJson	string	Array data collection marking allergen inclusions, e.g., ["Dairy"]
TagsJson	string	Search meta classifiers array string configurations, e.g., ["Keto"]

Table: NutritionFacts (Granular nutritional density splits)

Column Name	Type	Description / Example
Id	int	Primary Key (1-to-1 relationship mapping directly to Meals.Id)
Calories	int	Target absolute caloric balance, e.g., 650

Column Name	Type	Description / Example
Protein	double	Macro distribution parameter mass scale indicator, e.g., 55.0
Carbs	double	Macro distribution parameter mass scale indicator, e.g., 60.0
Fats	double	Macro distribution parameter mass scale indicator, e.g., 20.0
Fiber	double	Minor macro classification parsing property metric, e.g., 8.0

Table: Ingredients (Master physical asset items library)

Column Name	Type	Description / Example
Id	int	Primary Key
Name	string	Common taxonomy text element label, e.g., "Chicken Breast"

Table: MealIngredients (Join bridge linking ratios to menus)

Column Name	Type	Description / Example
Id	int	Primary Key
MealId	int	Foreign Key mapping to the target container row Meals.Id
IngredientId	int	Foreign Key linking the physical item Ingredients.Id
Amount	string	Quantity dimension descriptor layout value, e.g., "150g"

Table: MealPlanItems (Weekly calendar scheduling configurations)

Column Name	Type	Description / Example
Id	int	Primary Key
MealPlanId	int	Foreign Key mapping structural connections down to MealPlans.Id
MealId	int	Foreign Key linking specific components index Meals.Id
DayOfWeek	string	Scheduling placement marker constraints, e.g., "Monday"
MealTime	string	Target distribution window designation index, e.g., "Breakfast"

6. Smart Coach Service 🤖

Database Philosophy: Governs user dialog records and dashboard rendering data caches. It is highly transactional and supports localized document optimizations.

Table: ChatSessions (Dialogue grouping threads)

Column Name	Type	Description / Example
Id	int	Primary Key
UserId	int	Target account user reference link (Indexed Foreign Key)
StartedAt	DateTime	Initial implementation timeline milestone date
Title	string	Custom conversation categorization layout identifier, e.g., "Keto Planning"

Table: ChatMessages (Archives individual conversation line entries)

Column Name	Type	Description / Example
Id	int	Primary Key
ChatSessionId	int	Foreign Key tracking back directly to ChatSessions.Id
Role	string	Actor label context constraint value, e.g., "user" or "assistant"
Content	string	Raw dialogue text payload copy block
Timestamp	DateTime	Event persistence creation log time

Table: RecommendationCache (High-performance home screen dashboard caches)

Column Name	Type	Description / Example
UserId	int	System user primary identifier reference (Primary Key)
UserContextJson	string	Text representation mapping physical metadata snapshot states
HomeFeedDataJson	string	Full serialized data tree caching comprehensive dashboard lookups
CachedAt	DateTime	Real-time cache row compilation initialization timestamp
ExpiresAt	DateTime	Explicit expiration boundary (Standard 30-minute window limit)

7. Progress Tracking Service

Database Philosophy: This represents the write-heavy performance analytics engine. It functions as the permanent database tracking user activity metrics over time.

Table: WorkoutLogs (Routines completion metrics summaries)

Column Name	Type	Description / Example
Id	int	Primary Key
UserId	int	Target tracking actor identifier (Indexed Foreign Key)
WorkoutId	int	Unconnected template identifier mapping back to Workout service parameters
SessionId	string	Unique token tracker mapping to the runtime initialization instance
DurationInMinutes	int	Actual runtime spent by user executing the routine, e.g., 45
CaloriesBurned	int	Calculated energy metrics expenditure score, e.g., 350
Rating	int	Boundary verified structural score configuration, e.g., 5 (Scale 1–5)
Notes	string	User narrative input feedback data copy block (Nullable)
CompletedAt	DateTime	Processing execution end timestamp log

Table: WorkoutLogExercises (Set-by-set granular volume trackers)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
WorkoutLogId	int	Foreign Key tracking back directly to the parent layout WorkoutLogs.Id
ExerciseId	int	Unconnected movement template index identifier
SetsCompleted	int	Actual volume workload configuration tracking variables, e.g., 3
RepsCompleted	int	Actual iteration tracking variable completed, e.g., 12
WeightUsed	double	Resistance mass configuration used under tracking operations, e.g., 80.0
Completed	bool	Validation state flag confirming execution success loops

Table: WeightHistory (Chronological mass check logs)

Column Name	Type	Description / Example
Id	int	Primary Key
UserId	int	Target user index reference link (Indexed Foreign Key)
Weight	double	Numeric mass parameter indicator scale input value, e.g., 72.5
Date	DateTime	Day calendar timeline event registration point
Notes	string	Short description tracking note parameter (Nullable)

Table: BodyMeasurements (Physical perimeter scaling trackers)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
UserId	int	Target system user metric tracking link (Indexed)
Neck / Chest / Biceps / Waist / Hips / Thighs	double?	Dimensional metrics track inputs (Nullable strings for open tracking parameters)
RecordedAt	DateTime	Calendar tracking timeline event marking

Table: Achievements (Master milestone system rewards definitions)

Column Name	Type	Description / Example
Id	int	Primary Key
Name	string	Reward label text title string, e.g., "7-Day Streak"
Description	string	Qualification requirements summary copy block, e.g., "Worked out for 7 days..."
IconUrl	string	Image location file string reference, e.g., "/images/badges/streak7.png"

Table: UserAchievements (Join bridge tracking unlocked awards)

Column Name	Type	Description / Example
Id	int	Primary Key

Column Name	Type	Description / Example
UserId	int	Target account user reference link (Indexed Foreign Key)
AchievementId	int	Foreign Key mapping directly back to <code>Achievements.Id</code>
EarnedAt	DateTime	Realization milestone timeline tracking event

Table: Streaks (Monitors execution consistency loops)

Column Name	Type	Description / Example
UserId	int	Shared tracking profile key anchor (Primary Key)
CurrentStreak	int	Current consecutive calendar tracking interval counters accumulation, e.g., 8
LongestStreak	int	High-water mark consistency metric tracker preservation score, e.g., 21
LastWorkoutDate	DateTime?	Calendar reference used to analyze gap thresholds (Nullable)

Table: UserStatistics (High-speed cached query summaries repository)

Column Name	Type	Description / Example
Id	int	Primary Key (Must match <code>Users.Id</code> from Auth database)
TotalWorkouts	int	Aggregated cumulative metric count indicator index value, e.g., 46
TotalCaloriesBurned	int	Aggregated cumulative metric count indicator index value, e.g., 12850
CurrentWeight	double	Synced replication from newest active mass entries tracking, e.g., 72.5
StartWeight	double	Baseline profile capture weight entry parameter record tracking, e.g., 75.5
TotalWeightLost	double	Mathematically computed net progress scale tracking metric variance
UpdatedAt	DateTime	Profile row modification state processing timestamp

8. Subscription & Billing Service

Database Philosophy: The explicit system source of truth governing financial status flags, access tier authorizations, transactional metrics histories, and limit gate permissions.

Table: UserSubscriptions (Membership authorization tracking profiles)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
UserId	int	Context account owner reference tracker link (Unique Index)
Tier	string	Active classification boundary parameter choice (Free , Premium)
Status	string	System capability processing control state flag (Active , Expired , Cancelled)
StartsAt	DateTime	Initial execution timeline deployment date
ExpiresAt	DateTime	Absolute termination processing deadline checkpoint
UpdatedAt	DateTime	Record transaction index entry modification event

Table: BillingLogs (Audit payment receipt files logs)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
UserId	int	Associated user identity indexing criteria marker (Indexed)
ReferenceId	string	External mock transaction checkout confirmation unique code (VarChar(100))
Amount	decimal	Financial billing ledger transaction metric formatting data scale (199.99)
Currency	string	ISO localization financial ticker definition, e.g., "EGP"
PaymentStatus	string	Log confirmation status execution results values (Success , Failed)
Timestamp	DateTime	Transaction logging processing milestone

9. Notification Service

Database Philosophy: Functions as a decoupled tracking cache system archiving alert records and routing parameters for inside-app messaging lists.

Table: InAppNotifications (App layout messaging feeds entries logs)

Column Name	Type	Description / Example
Id	int	Primary Key (Identity)
UserId	int	Target notification distribution destination identifier (Indexed)
Title	string	Notification heading copy block text identifier string, e.g., "7-Day Streak!"
Message	string	Detailed content instruction narrative copy string, e.g., "Worked out 7 consecutive days..."
Type	string	Category tracking identifier flags mapping (AchievementAlert , StreakWarning)
IsRead	bool	Boolean interaction check tracking state flag (Defaults to false)
SentAt	DateTime	System delivery network outbound timestamp logging